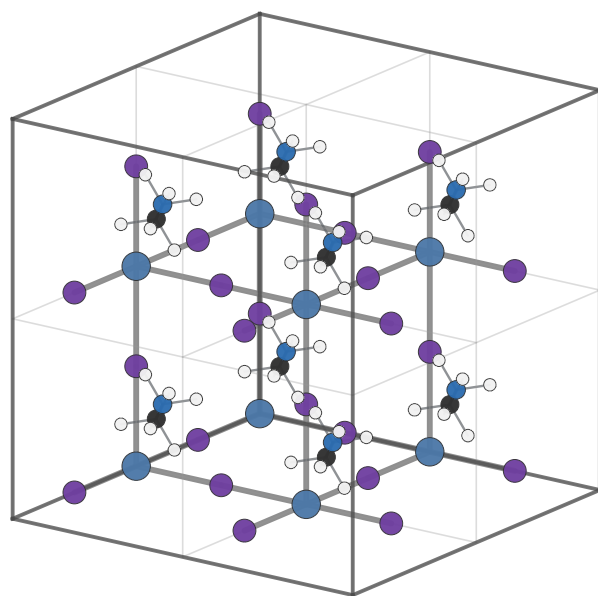


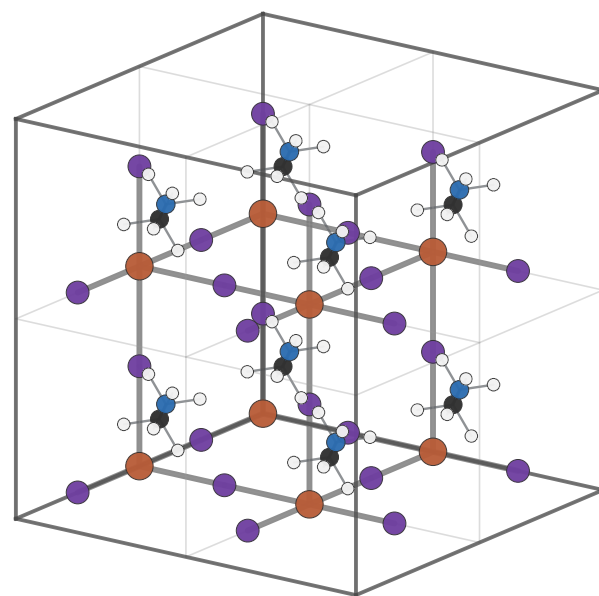
(a)

MAPbI₃



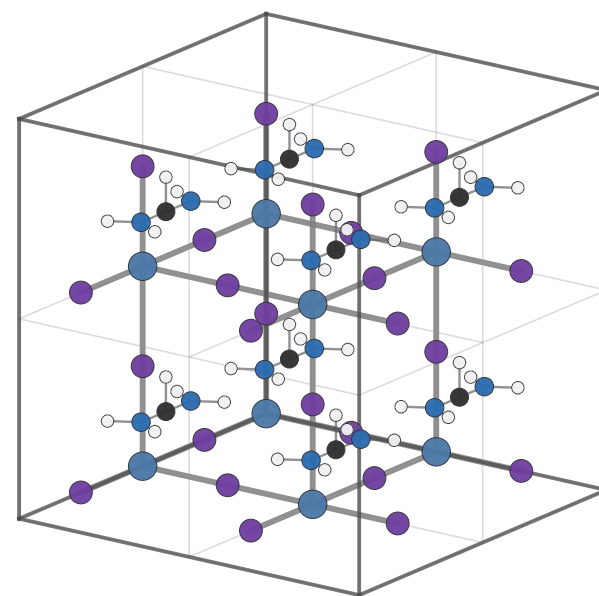
(b)

MASnI₃



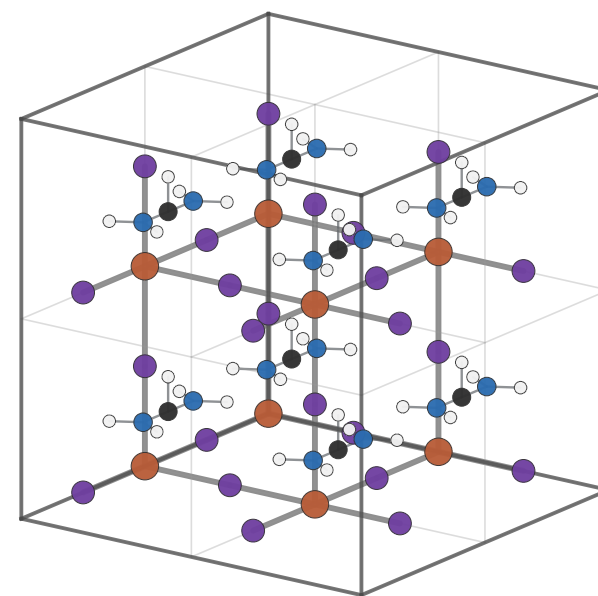
(c)

FAPbI₃



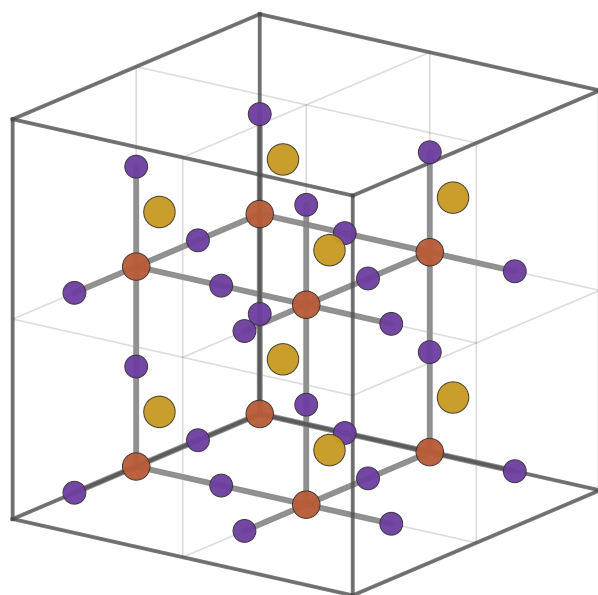
(d)

FASnI₃



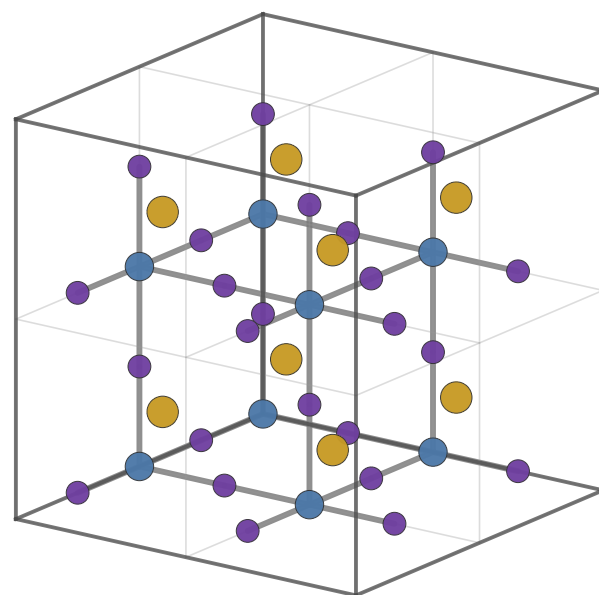
(e)

CsSnI₃



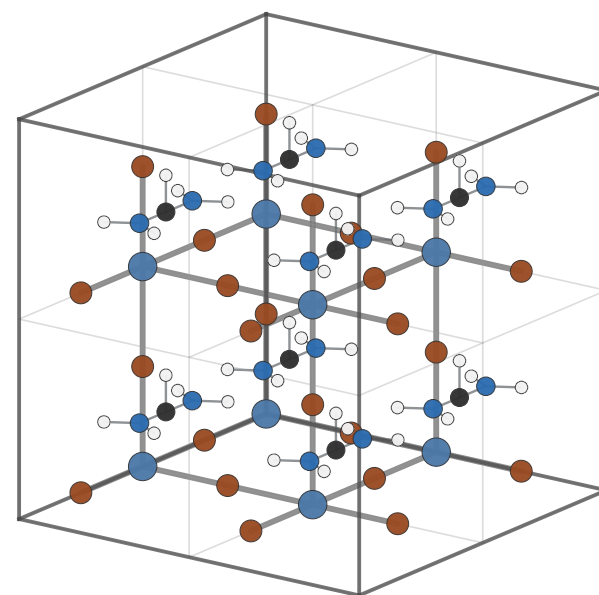
(f)

CsPbI₃



(g)

FAPbBr₃



(h)

FASnBr₃

